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Reusable minimally invasive surgical instrument

Abstract

A reusable tip for minimally invasive surgical instruments can include an end effector, a two-part hub, and a yoke. The end effector can include one or more movable portions enabling the end effector to move between first (e.g., open) and second (e.g., closed) positions via manipulation of the yoke within the hub. The hub can include a proximal hub and a distal hub coupled together. The end effector can be coupled to the distal hub, which can be made of a metal to provide high strength and secure shape to the distal hub. The proximal hub can be made of an autoclavable insulation material capable of maintaining a removable coupling with a control shaft. The proximal hub can be made of polyetheretherketone (PEEK).

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/298,817, filed Mar. 11, 2019, now U.S. Pat. No. 11,813,017, issued Nov. 14, 2023, which is hereby incorporated by reference for all purposes.

TECHNICAL FIELD

(1) The present disclosure relates to surgical equipment generally and more specifically to reusable tips for minimally invasive surgical instruments, such as laparoscopic probes.

BACKGROUND

(2) Minimally invasive surgical techniques can involve inserting instruments into a body cavity through existing openings or small incisions to perform surgical interventions. The surgical instruments used in such techniques are manipulated remotely, such as via robotic control or human control. For example, in endoscopic or laparoscopic surgeries, an individual may manipulate a surgical instrument within a patient using handles connected to the surgical instrument by control shafts passing into the patient. Thus, surgical interventions can be conducted with less damage to

- the patient, shorter healing time, and less risk of infection as compared to traditional surgery.
- (3) Often, minimally invasive surgical techniques involve the use of electrosurgical instruments. Electrosurgical instruments use the application of electrical currents to perform various surgical tasks, such as cutting, coagulation, desiccation, or fulguration of tissue. Electrical currents can pass through the control shaft and into tissue surrounding the end effector of the surgical instrument.
- (4) Many minimally invasive surgical instruments are made to be one-time use instruments. Thus, each instrument is manufactured to be a sterile instrument that, after use, is not intended to be or cannot be re-sterilized, and is discarded.
- (5) Many minimally invasive surgical instruments are designed to have a reusable handle to which different shafts can be attached. Each shaft can have a different type of end effector, such as scissors or graspers. In some cases, the entire shaft can be removed and sterilized. However, due to the length of the shafts used in various minimally invasive surgeries, these tools can be burdensome to sterilize, such as requiring custom or extra-large sterilization pouches or occupying so much volume in an autoclave that a single set of instruments may need to be autoclaved in multiple batches. As a result, current sterilizable minimally invasive surgical instruments can take a very long time to sterilize and can require the expenditure of multiple batches worth of energy and resources.
- (6) Some single-use, disposable surgical tips have been used, comprising a one-piece hub that is partially covered by a section of polyolefin heat-shrink tubing. However, such surgical tips could not be sufficiently cleaned and sterilized for reuse. Attempts to clean and sterilize such tips resulted in loose heat-shrink tubing that carried risk of detachment during use, which could lead to inadvertent electrical conduction to adjacent tissue, loss of the tubing within a patient, and malfunction of the surgical tip due to mechanical interference from the loose tubing. As a result, surgical interventions using such tips required disposing of the tips after use, resulted in large amounts of waste and requiring institutes to maintain large inventories of new tips, thus occupying critical space and increasing costs.

SUMMARY

- (7) The term embodiment and like terms are intended to refer broadly to all of the subject matter of this disclosure and the claims below. Statements containing these terms should be understood not to limit the subject matter described herein or to limit the meaning or scope of the claims below. Embodiments of the present disclosure covered herein are defined by the claims below, not this summary. This summary is a high-level overview of various aspects of the disclosure and introduces some of the concepts that are further described in the Detailed Description section below. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this disclosure, any or all drawings and each claim.
- (8) Embodiments of the present disclosure include is a surgical tip comprising: an end effector movable from a first position to a second position, the end effector having a distal region for contacting tissue and a proximal region; a yoke coupled to the proximal region of the end effector for manipulating the end effector between the first position and the second position, wherein the yoke comprises a threaded proximal end, wherein the yoke and the end effector are made of an electrically conductive material, and wherein the yoke and the end effector are electrically coupled; and a hub positioned around the yoke, wherein the hub comprises a distal hub and a proximal hub, wherein the end effector is coupled to the distal hub at a pivot, wherein the proximal hub is made of a non-conductive material, and wherein the proximal hub comprises a proximal threaded end.
- (9) In some cases, the proximal hub is made of Polyetheretherketone. In some cases, the proximal hub is coupled to the distal hub by a threaded connection. In some cases, the proximal hub is coupled to the distal hub with an epoxy. In some cases, the threaded proximal end of the yoke and the threaded proximal end of the proximal hub are coaxially threaded. In some cases, the proximal

hub surrounds a portion of the distal hub. In some cases, the end effector is selected from the group consisting of scissors, graspers, a punch, and dissectors. In some cases, the distal hub includes a slot for receiving the proximal region of the end effector during movement of the end effector between the first position and the second position, and wherein the proximal hub surrounds a portion of the distal hub located proximally beyond the slot. In some cases, the proximal hub is made of material having a dielectric strength of at or greater than 130 KV/cm. In some cases, the proximal hub has an outer diameter of at or less than 6.0 mm. In some cases, the proximal hub is made of an autoclavable and sterilizable material. In some cases, the proximal hub is made of a material having a glass transition temperature at or above 120° C. In some cases, a length between a distal end of the distal hub and a proximal end of the yoke is at or less than 5 cm. In some cases, a length between a distal end of the end effector and a proximal end of the yoke is at or less than 9 cm. In some cases, the end effector comprises a stationary portion and a movable portion, wherein the stationary portion is fixed with respect to the hub. In some cases, the end effector comprises a first movable portion and a second movable portion, wherein the first movable portion and the second movable portion move with respect to the hub during movement of the end effector between the first position and the second position.

(10) Embodiments of the present disclosure include a surgical apparatus comprising: a surgical tip as described above; and a handpiece couplable to the surgical tip, wherein the handpiece comprises an inner shaft movable axially within an outer shaft, wherein the outer shaft is couplable to the proximal hub by the proximal threaded end of the proximal hub, and wherein the inner shaft is couplable to the yoke by the threaded proximal end of the yoke.

(11) In some cases, the outer shaft is made of the same material as the proximal hub. In some cases, the outer shaft has an outer diameter that is the same size as an outer diameter of the proximal hub. In some cases, the handpiece further comprises a cautery pin electrically coupled to the inner shaft to convey electrical signals through the inner shaft and the yoke to the end effector when the surgical tip is coupled to the handpiece. In some cases, the surgical apparatus further comprises an additional surgical tip, wherein the handpiece is couplable to the additional surgical tip after decoupling of the surgical tip from the handpiece, wherein the additional surgical tip comprises an additional end effector that is different from the end effector of the surgical tip.

(12) Embodiments of the present disclosure include a method, comprising: providing a surgical tip as described above; coupling the surgical tip to a handpiece, wherein the handpiece comprises an inner shaft movable axially within an outer shaft, and wherein coupling the surgical tip to the handpiece comprises coupling the outer shaft of the handpiece to the proximal hub of the surgical tip by the proximal threaded end of the proximal hub, and coupling the inner shaft of the handpiece to the yoke of the surgical tip by the threaded proximal end of the yoke; decoupling the surgical tip from the handpiece; and autoclaving the surgical tip.

(13) In some cases, the method further comprises coupling the autoclaved surgical tip to the handpiece or an additional handpiece; and contacting tissue with the end effector of the surgical tip during a surgical procedure. In some cases, the method further comprises applying electrical current to tissue through the yoke and the end effector. In some cases, the electrical current conveyed through the yoke is not conveyed through the proximal hub.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The specification makes reference to the following appended figures, in which use of like reference numerals in different figures is intended to illustrate like or analogous components.

(2) FIG. 1 is a set of schematic side views depicting surgical instrument tips in closed and open positions and a partial-cutaway schematic side view of a distal portion of a control shaft according

to certain aspects of the present disclosure.

(3) FIG. 2 is a set of schematic side views depicting surgical instrument tips coupled to control shafts in closed and open positions according to certain aspects of the present disclosure.

(4) FIG. 3 is a schematic side view depicting a minimally invasive surgical instrument comprising a surgical instrument tip installed on a control shaft coupled to a handle according to certain aspects of the present disclosure.

(5) FIG. 4 is an exploded schematic side view depicting a surgical instrument tip according to certain aspects of the present disclosure.

(6) FIG. 5 is a cross-sectional top view depicting a hub of a surgical instrument tip with the proximal hub and distal hub coupled together according to certain aspects of the present disclosure.

(7) FIG. 6 is a cross-sectional top view depicting a hub of a surgical instrument tip with the proximal hub and distal hub separate according to certain aspects of the present disclosure.

(8) FIG. 7 is a flowchart depicting a process for using a surgical instrument tip according to certain aspects of the present disclosure.

(9) FIG. 8 is a set of schematic side views depicting surgical instrument tips with flared proximal hubs in closed and open positions according to certain aspects of the present disclosure.

DETAILED DESCRIPTION

(10) Certain aspects and features of the present disclosure relate to reusable tips for minimally invasive surgical instruments. The reusable tip can include an end effector, a hub, and a yoke. The end effector can include one or more movable portions enabling the end effector to move between a first (e.g., open) position and a second (e.g., closed) position via manipulation (e.g., axial manipulation) of the yoke within the hub. The hub can include a proximal hub and a distal hub that are coupled together, such as via a threaded coupling and/or epoxy. The end effector can be coupled to the distal hub, which can be made of a metal to provide high strength and secure shape to the distal hub. The proximal hub can be made of an autoclavable insulation material capable of maintaining a removable coupling with a control shaft. The proximal hub can be made of polyetheretherketone (PEEK).

(11) Certain aspects of the present disclosure relate to reusable tips for minimally invasive surgical instruments. These surgical tips can be configured for any suitable type of surgery, such as robot-conducted or robot-assisted surgeries, endoscopic surgeries, laparoscopic surgeries, or any other suitable minimally invasive surgery. In some cases, certain aspects of the present disclosure can be especially beneficial for laparoscopic surgical instruments due to the style of surgical instruments used in laparoscopic surgeries.

(12) A minimally invasive surgical instrument can include a surgical tip removably attachable to a handle via a control shaft. The handle can be non-removably coupled to the control shaft, although that need not always be the case. The combination of a handle and a control shaft, whether the control shaft is removable or not, can be known as a handpiece. The surgical tip is removably couplable to the control shaft, permitting the surgical tip to be cleaned and sanitized on its own. Further, the functionality of the surgical instrument can be quickly changed on the fly, such as mid-surgery, by removing one surgical tip and attaching a different surgical tip. In this fashion, a single handle and control shaft can be used with multiple surgical tips during a surgical procedure, resulting in only a single handle and control shaft set needing to be cleaned and sterilized despite the use of multiple different surgical tips with that same handle and control shaft. As used herein, the surgical instrument can be referred to as having the handle at its proximal end and the surgical tip at its distal end. Thus, the terms distal and proximal as used herein can refer to directions away from and towards the handle-end of the surgical instrument.

(13) The handle can include one or more grips for actuating the end effector of the surgical tip. The handle can be configured in different fashions depending on the desired functionality. Some handles can have ratcheting action (e.g., ratcheting grips) while others do not. In some cases, handles can include one or more electrical posts (e.g., cautery posts) to convey electrical current to

the control shaft. Some handles can have short electrical posts while others have long electrical posts. In some cases, a handle can include a flush port to facilitate cleaning of the handle and/or attached control shaft. In some cases, the electrical post can double as a flush port. In some cases, a rotation knob can control rotation of the control shaft, which can in turn control rotation of the end effector of the surgical tip.

(14) In some cases, the control shaft can be permanently attached to the handle, although that need not always be the case. The control shaft can be made of an outer shaft and an inner shaft. The outer shaft can be made of an electrically insulating material, such as PEEK. The inner shaft can be moveable within the outer shaft, such as axially movable, to enable actuation of the end effector via manipulation of the inner shaft with respect to the outer shaft. The inner shaft can be made of an electrically conductive material, such as a metal. In some cases, the inner shaft can be made of the same material as the yoke, end effector, and/or distal hub of the surgical tip.

(15) Surgical tips disclosed herein can make use of any suitable style of end effector. Suitable styles of end effectors include scissors, graspers, punches, and dissectors. As an example, suitable scissor configurations can include curved scissors, straight scissors, Metzenbaum-style scissors, hooked scissors, and the like. As an example, suitable grasper configurations can include atraumatic graspers, fenestrated graspers, clinching graspers, Babcock-style graspers, Hunter-style graspers, Allis-style graspers, and the like. As an example, suitable punches can include cupped punches, biopsy punches, and the like. As an example, suitable dissectors can include dolphin-nose-style dissectors, Maryland-style dissectors, Birkett grasper-style dissectors, and the like.

(16) End effectors can include one or more movable portions. In some cases, an end effector can move between first and second positions by moving a single movable portion against a stationary portion, such as in a common biopsy punch. In the example of a biopsy punch, a movable portion (e.g., blade) can be forced against a stationary portion (e.g., a biopsy collection surface) to collect a biopsy. This type of action with a single movable portion can be known as single-action, and an end effector using this type of action can be known as a single-action end effector. In some cases, an end effector can have two movable portions, often opposing one another, such as in a common pair of scissors. In the example of scissors, two movable portions (e.g., scissor blades) can be moved towards one another to initiate cutting action. This type of action with two movable portions can be known as dual-action, and an end effector using this type of action can be known as a dual-action end effector. Aspects of the present disclosure can be used with single-action or dual-action end effectors. In some cases, aspects of the present disclosure can be used with no-action end effectors having no movable portions (e.g., electrosurgical electrodes).

(17) Each surgical tip can include an end effector coupled to a hub and a yoke. The end effector can be coupled to the hub such that movement of the yoke within the hub causes the end effector to move between first and second positions. For example, the end effector can be secured to the hub by way of a rivet, with a shaft of the yoke engaging one or more slots of the end effector such that axial movement of the yoke causes one or more portions of the end effector to pivot about the rivet. In some cases, other configurations can be used. The end effector can be coupled to the hub such that it is axially fixed with respect to the hub but one or more movable portions of the end effector are rotatably movable with respect to the hub.

(18) The yoke can be positioned within the hub, within an inner diameter of the hub. The yoke can be coupled to the end effector in any suitable fashion such that movement of the yoke with respect to the hub causes movement of the one or more movable portions of the end effector with respect to the hub. In some cases, axial movement of the yoke is translated into rotational movement of one or more movable portions of the end effector about a pivot, although this need not always be the case. In some cases, axial movement of the yoke can affect the end effector in other ways. In some cases, rotational movement of the yoke can affect the end effector. The yoke can be coupled to the one or more movable portions of the end effector, such as using a sliding linkage coupling (e.g., a pin and slot linkage). The yoke can couple to the end effector using a sliding linkage coupling or

any suitable mechanical coupling for converting axial movement of the yoke into appropriate movement of the movable portion(s) of the end effector. For example, a yoke can include a shaft or pin that fits within a slot on a movable portion of an end effector such that movement of the yoke in an axial direction forces the movable portion in an upwards or downwards direction, causing the movable portion to pivot about a pivot axis. The yoke can couple to the control shaft (e.g., an inner shaft of the control shaft) through any suitable technique. In some cases, the yoke can couple to the control shaft via a threaded connection.

(19) The hub is a multi-part hub comprising a proximal portion and a distal portion, which can be referred to as a proximal hub and a distal hub, respectively. The distal hub can be used to fixedly or movably secure portions of the end effector. For example, a single-action end effector can include a stationary portion that is fixed with respect to the distal hub and a movable portion that is movably coupled to the distal hub. In another example, a dual-action end effector can include two movable portions that are movably coupled to the distal hub. In some cases, one or more movable portions of the end effector can be movably coupled to the distal hub via rotatable coupling, such as a rivet, screw, or bolt. The distal end can serve as a fulcrum for the one or more movable portions of the end effector. In some cases, the end effector can be coupled to the distal hub to act as a class one lever (e.g., a single class one lever or a double class one lever). In some cases, the end effector can be coupled to the distal hub to act as a class three lever.

(20) The distal hub can be made of a material having sufficient strength and/or hardness to withstand construction of the end effector (e.g., application of a rivet when a rivet is used) and repeated actuation of the end effector. In some cases, the distal hub can be made of a metal suitable for surgical tools, such as surgical stainless steel or the like. In some cases, the distal hub is made of a conductive material, although that need not always be the case. In some cases, the distal hub is made of metal to ensure repeated actuation of the end effector would not negatively impact the coupling between the end effector and the distal hub for a designed lifetime of the surgical tip. In some cases, the distal hub is made from the same material as the end effector. In some cases, the distal hub is made from the same material as the one or more movable portions of the end effector.

(21) The proximal hub can be removably or permanently coupled to the distal hub. The proximal hub can couple the distal hub to the control shaft (e.g., to an outer shaft of the control shaft). The proximal hub can include a distal coupling portion configured to couple to the distal hub, and a proximal coupling portion configured to couple to the control shaft. The distal coupling portion can be a threaded connection, although that need not always be the case. In an example, the distal hub and proximal hub can be coupled together via a threaded connection. In some cases, the distal hub and proximal hub can be coupled together with an adhesive material, such as an epoxy. In some cases, the distal hub and proximal hub can be coupled together using multiple techniques, such as via a threaded connection and an epoxy.

(22) In some cases, all portion(s) of an end effector that are enclosed by the distal hub can remain enclosed by the distal hub (e.g., remain within the circumference of the distal hub) throughout actuation of the end effector between first and second positions. For example, while a pair of scissors may have blades that can open further than the circumference of the distal hub, the proximal ends of the movable portions of the end effector can be configured to remain within the circumference of the distal hub during actuation. However, in some cases, actuation of the end effector between first and second positions can cause one or more proximal ends of the end effector to extend outside of the circumference of the distal hub, such as through a slot in a wall of the distal hub. In an example, certain graspers operating as class one levers may have proximal ends that extend outwards from the circumference of the distal hub through slots to permit the jaws of the grasper to fully open.

(23) The proximal hub can act to couple the distal hub to the control shaft while also acting as an insulator. The proximal hub can be made of an insulating material (e.g., an insulator). The proximal hub can be made of a material having a sufficiently high dielectric strength. In some cases, the

dielectric strength of the proximal hub can be at or greater than approximately 230 KV/cm, such as at or greater than approximately 130 KV/cm, 135 KV/cm, 140 KV/cm, 145 KV/cm, 150 KV/cm, 155 KV/cm, 160 KV/cm, 165 KV/cm, 170 KV/cm, 175 KV/cm, 180 KV/cm, 185 KV/cm, 190 KV/cm, 195 KV/cm, 200 KV/cm, 205 KV/cm, 210 KV/cm, 215 KV/cm, 220 KV/cm, 225 KV/cm, 230 KV/cm, 235 KV/cm, 240 KV/cm, 245 KV/cm, and/or 250 KV/cm.

(24) Generally, the surgical tip is made of metal parts except for the proximal hub. For example, the yoke, end effector, and distal hub can be made of a metal, which can be the same type of metal. Since these metal parts are generally electrically conductive, the proximal hub can be made from an electrically insulating material to protect non-targeted tissue surrounding the surgical tip during use. The proximal hub can act to insulate the yoke and at least a portion of the distal hub from surrounding tissue, such as during a surgical procedure. Since the yoke can convey electrical signals to the end effector in some surgical procedures, protection to surrounding tissue can be important to reduce the risk of inadvertent damage to non-targeted tissue. Non-targeted tissue can include any tissue surrounding the surgical tip during use to which electrical current is not desired to be directed, such as tissue not in contact with the end effector. The proximal hub can surround a portion of the distal hub. In some cases, the proximal hub can surround at least 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, and/or 100% of the length of the distal hub. In some cases, this portion of the distal hub that is surrounded can be measured from a proximal end of the distal hub. In cases where the distal hub comprises a slot or other opening, such as to pass one or more proximal ends of an end effector through during actuation of the end effector, the proximal hub can extend from the proximal end of the distal hub up to or approximately up to the slot or other opening. In some cases, such as if proximal end(s) of the end effector remain entirely within the distal hub during actuation of the end effector, the proximal hub can extend up to or approximately up to the distal end of the distal hub.

(25) In some cases, the proximal hub is made from a thermoplastic polymer that is suitable for use in surgical tools and is sterilizable. In some cases, the proximal hub is made from PEEK. In some cases, the use of PEEK can enable the proximal hub to resist damage from inadvertent contact with other tools and surfaces. In some cases, a PEEK proximal hub can resist damage better than a heat-shrink insulator surrounding a single-piece hub. The proximal hub can be made from a material having a glass transition temperature that is at or above 121° C. or 132° C. (e.g., at or above 140° C., or 143° C.). In some cases, the proximal hub can be made from a material having a melting temperature that is at or greater than 300° C. (e.g., at or above 340° C. or 343° C.). The sufficiently high glass transition temperature and/or melting temperature can ensure the proximal hub does not lose shape or integrity when subjected to sterilization procedures in an autoclave, which can reach temperatures of up to 121° C. (e.g., for gravity-based sterilizers) or 135° C. (e.g., for vacuum-based sterilizers). In some cases, the proximal hub can be made of a material having a glass transition temperature above a maximum temperature of a sterilization procedure for sterilizing the surgical tip. Such a sterilization procedure can be a sterilization procedure shared with other equipment commonly used with the surgical tip, such as a scalpel used to create an incision into which the surgical tip can be maneuvered.

(26) The proximal hub can be coupled to the control shaft through any suitable technique. In some cases, the proximal hub is coupled to the control shaft via a threaded connection. To ensure a reliable connection between the proximal hub and the control shaft, the proximal hub can be made of a material capable of retaining any coupling features used to couple the proximal hub to the control shaft, such as threads. In some cases, the proximal hub can contain threads on an inner diameter of the proximal hub, which engage threads on an outer diameter of a portion of the outer shaft of the control shaft, however this need not always be the case. In some cases, the threaded connection between the proximal hub and the control shaft and the threaded connection between the yoke and the control shaft are coaxial.

(27) The entire surgical tip can be of any suitable dimensions. In some cases, a surgical tip for

minimally invasive surgeries can be at or less than 9 cm in length from the distal end of the end effector to the proximal end of the yoke. In some cases, the surgical tip can be at or less than 9, 8.5, 8, 7.5, 7, 6.5, or 6 cm from the distal end of the end effector to the proximal end of the yoke. In some cases, a surgical tip for minimally invasive surgeries can be at or less than 5 cm in length from the distal end of the distal hub to the proximal end of the yoke. In some cases, the surgical tip can be at or less than 6.5, 6, 5.5, 5, 4.5, 4, 3.5, or 3 mm from the distal end of the distal hub to the proximal end of the yoke. In some cases, the surgical tip can have a length that is at or less than approximately 40%, 38%, 36%, 34%, 32%, 30%, 28%, 26%, 24%, 22%, 20%, 18%, 16%, 14%, 12%, 10%, 8%, 6%, and/or 4% of the length of a control shaft coupled to the surgical tip. In some cases, the surgical tip can have a diameter that is at or less than 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, or 5 mm in diameter. In some cases, the hub can have an outer diameter that is at or less than 12, 11, 10, 9, 8, 7, 6, or 5 mm in diameter.

(28) In some cases, the proximal hub can have an outer diameter that is the same as an outer diameter of the control shaft (e.g., outer diameter of an outer shaft of the control shaft). In some cases, the proximal hub can have an outer diameter that is constant throughout its length. However, in some cases, the outer diameter of the proximal hub can become larger and optionally also smaller extending distally from a proximal end of the proximal hub. In some cases, a proximal hub with a flared diameter can provide a contiguous outer surface at the coupling between the proximal hub and the control shaft, while also providing a thicker proximal hub over portions of the distal hub, such as to provide a stronger insulating effect.

(29) The entire surgical tip disclosed herein can be cleanable and sterilizable. In some cases, the entire surgical tip can be sterilizable through the use of an autoclave (e.g., autoclavable). The materials and construction of the components of the surgical tip can be suitable for standard operating room cleaning and sterilizing procedures.

(30) The ability to remove the relatively small surgical tip from the control shaft and handle permit the relatively small surgical tip to be easily handled during any initial cleaning steps, and thereafter easily sterilized in bulk with other equipment. A single surgical tip as disclosed herein can fit within relatively small sterilization pouches, permitting numerous surgical tips to be sterilized at once. By contrast, sterilizable laparoscopic equipment comprising a tip and non-removable control shaft (e.g., larger than 9 or 10 cm in overall length) may require the use of large and/or customized sterilization pouches, and may occupy so much room in an autoclave that one or only a few items can be sterilized at a time.

(31) Further, the relatively small size of the surgical tip disclosed herein can permit multiple different styles of surgical tip to be stored in a relatively small volume, at least as compared to sterilizable laparoscopic equipment comprising a tip and non-removable control shafts. For example, numerous surgical tips as disclosed herein can be stored in a volume (e.g., in a container) without needing to store a control shaft for each tip. Rather, a single handle and control shaft can be stored, to which any desired surgical tip can be attached.

(32) Additionally, due to the modular nature of certain aspects of the disclosure, any number of tips can be combined with any number of handles and/or control shafts to achieve the desired surgical instrument for a particular surgery. For example, different handle options can include handles with or without ratcheting grips, with different sized electrical posts, and with different style rotation knobs. Further, different control shaft options can include different length control shafts (e.g., 25 cm, 34 cm, 42 cm, or other lengths). Thus, to provide a wide array of surgical instruments, aspects of the present disclosure can provide a set of surgical tips and a set of handles and/or control shafts which can be mixed and matched as necessary to achieve the desired instrument. By contrast, standard laparoscopic tools with surgical tips non-removably attached to control shafts require a new tip for each and every potential instrument length for each and every potential type of instrument, thus requiring the storage of many more items that occupy larger volumes of space than the surgical tips according to certain aspects of the present disclosure.

(33) Aspects and features of the present disclosure enable minimally invasive surgical equipment that is easy and efficient to store, is easy to use, has well-insulated surgical tips, can be easily and efficiently sterilized, and is highly customizable.

(34) These illustrative examples are given to introduce the reader to the general subject matter discussed here and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements, and directional descriptions are used to describe the illustrative embodiments but, like the illustrative embodiments, should not be used to limit the present disclosure. The elements included in the illustrations herein may not be drawn to scale.

(35) FIG. 1 is a set of schematic side views depicting surgical instrument tips **100**, **102**, in closed and open positions and a partial-cutaway schematic side view of a distal portion of a control shaft **104** according to certain aspects of the present disclosure. Surgical tip **100** can comprise an end effector **106**, a multi-part hub **108**, and a yoke **110**. The yoke **110** is positioned within hub **108** and axially movable with respect to the hub **108** to actuate the end effector **106**. Surgical tip **100** can have an end effector **106** in a first (e.g., closed) position. Surgical tip **102** can be surgical tip **100** with the end effector **106** in a second (e.g., open) position.

(36) The end effector **106** can include one or more movable portions. As depicted in FIG. 1, the end effector **106** includes a first movable portion **116** and a second movable portion **118**. The first and second movable portions **116**, **118** are rotatable about a pivot **120** secured to hub **108**. The pivot **120** can be secured to the distal hub **112** of the multi-part hub **108**. Thus, the end effector **106** can be coupled to the distal hub **112** of the multi-part hub **108**. The first and second movable portions **116**, **118** can be movable coupled with respect to the yoke **110** to permit the first and second movable portions **116**, **118** to move between a first position as depicted with respect to surgical tip **100** and a second position as depicted with respect to surgical tip **102** when the yoke **110** is moved axially with respect to the hub **108**. For example, movement of the yoke **110** in direction **140** can cause the first and second movable portions **116**, **118** to move in inward directions **134** to close the first and second movable portions **116**, **118** against one another, while movement of the yoke **110** in direction **142** can cause the first and second movable portions **116**, **118** to move in outward directions **136** to open and separate the first and second movable portions **116**, **118** from one another. Movement of the yoke **110** over distance **144** can represent actuation of the end effector **106** between a first position and a second position. As depicted in FIG. 1, the first and second movable portions **116**, **118** are jaws of a grasper, although other movable portions can be used.

(37) In some cases, a movable portion of an end effector **106** can have a proximal end that sticks out beyond a circumference of the distal hub **112** when in the first or second position. As depicted in FIG. 1, the first and second movable portions **116**, **118** of surgical tip **102** have proximal ends **138** that move along a path during actuation of the end effector **106**, which path takes the proximal ends **138** outside of the circumference of the distal hub **112**. The distal hub **112** can include one or more slots through which the proximal ends **138** may pass. The proximal hub **114** can overlap a portion of the distal hub **112** up to or approximately up to such slot(s). In some cases, such as if no slot exists because the proximal ends of the end effector remain within the circumference of the distal hub during actuation, the proximal hub can overlap a portion of the distal hub that extends for up to or approximately up to the pivot **120** or up to or approximately up to the distal end of the distal hub.

(38) The multi-part hub **108** can comprise a distal hub **112** coupled to a proximal hub **114**. While the distal hub **112** can be made of a metal, such as the same metal used for the end effector **106** and/or the yoke **110**, the proximal hub **114** can be made of an insulator, such as PEEK. The proximal hub **114** can overlap and surround a portion of the distal hub **112**, although that need not always be the case.

(39) The surgical tip **100** can removably couple to a control shaft **104**. The control shaft **104** can include an inner shaft **128** and an outer shaft **126**. For illustrative purposes, the outer shaft **126** is

depicted in a partial-cutaway view and the inner shaft **128** is shown in a cutaway side view.

(40) The inner shaft **128** can be axially movable within the outer shaft **126** to facilitate actuation of the end effector **106** via axially moving the yoke **110**. The inner shaft **128**, as well as the yoke **110** and end effector **106**, can be made of electrically conductive materials, such as electrically conductive metal. The outer shaft **126** can be made of an electrically insulating material, such as PEEK. The outer shaft **126** can be made of the same material as the proximal hub **114**.

(41) The proximal hub **114** can removably couple to the outer shaft **126** of the control shaft **104** and the yoke **110** can removably couple to the inner shaft **128** of the control shaft **104**. The proximal hub **114** can include threads **122** (e.g., internal threads) that engage threads **130** (e.g., external threads) of the outer shaft **126**. The yoke **110** can include threads **124** (e.g., external threads) that engage threads **132** (e.g., internal threads) of the inner shaft **128**. For illustrative purposes, threads **122** are depicted in dotted line for surgical tip **100**. In some cases, threads **122**, **124** are coaxial. Likewise, threads **130**, **132** can be coaxial.

(42) As used herein with regards to FIG. **1** and other suitable figures, movement of an inner shaft **128** or movement of a yoke **110** refers to movement of the inner shaft **128** or yoke **110** with respect to the outer shaft **126** or hub **108**, respectively. Because the hub **108** is axially fixed with respect to the outer shaft **126** via a coupling (e.g., threaded coupling) between the outer shaft **126** and the proximal hub **114**, and because the inner shaft **128** is axially fixed to the yoke **110** via a coupling (e.g., threaded coupling), movement of the inner shaft **128** axially with respect to the outer shaft **126** causes the yoke **110** to move axially with respect to hub **108**, thus actuating the end effector to move between first and second positions.

(43) FIG. **2** is a set of schematic side views depicting surgical instrument tips **200**, **202** coupled to control shafts **204** in closed and open positions according to certain aspects of the present disclosure. Surgical tip **200** can comprise an end effector **206**, a multi-part hub **208**, and a yoke. The yoke is positioned within hub **208** and axially movable with respect to the hub **208** to actuate the end effector **206**. Surgical tip **200** can have an end effector **206** in a first (e.g., closed) position. Surgical tip **202** can be surgical tip **200** with the end effector **206** in a second (e.g., open) position. Surgical tip **200** and control shaft **204** can be surgical tip **100** and control shaft **104** of FIG. **1**.

(44) As depicted in FIG. **2**, the surgical tip **200**, **202** is coupled to the control shaft **204**. The proximal hub **214** of the multi-part hub **208** is coupled (e.g., threadably coupled) to the outer shaft **226** of the control shaft **204**. The yoke is coupled (e.g., threadably coupled) to the inner shaft **228** of the control shaft **204**. Thus, movement of the inner shaft **228** in direction **240** can cause the first and second movable portions **216**, **218** to move in inward directions **234** to close the first and second movable portions **216**, **218** against one another, while movement of the yoke **210** in direction **242** can cause the first and second movable portions **216**, **218** to move in outward directions **236** to open and separate the first and second movable portions **216**, **218** from one another. The inner shaft **228** can move over distance **244**, which can represent actuation of the end effector **206** between a first position and a second position. As depicted in FIG. **2**, the first and second movable portions **216**, **218** are jaws of a grasper, although other movable portions can be used.

(45) FIG. **3** is a schematic side view depicting a minimally invasive surgical instrument **300** comprising a surgical instrument tip **302** installed on a control shaft **304** coupled to a handle **346** according to certain aspects of the present disclosure. The surgical instrument **300** can include a surgical tip **302** (e.g., surgical tip **102** of FIG. **1**), a control shaft **304** (e.g., control shaft **104** of FIG. **1**), and a handle **346**. The surgical tip **302** is removably coupled to the control shaft **304**. In some cases, the control shaft **304** can be removably coupled to the handle **346**, although that need not be the case. In some cases, the control shaft **304** is permanently coupled to the handle **346**.

(46) The surgical tip **302** can include a multi-part hub comprising a distal hub **312** and a proximal hub **314**. The surgical tip **302** can be removably coupled to the control shaft **304** via proximal hub **314**, which is in turn permanently coupled to distal hub **312**, which is movably secured to the end

effector **306**.

(47) The handle **346** can include one or more grips **352** for manipulating the end effector **306** of the surgical tip **302**. Movement of the one or more grips **352** (e.g., movement with respect to one another, such as towards or away from one another) can cause the inner shaft of the control shaft **304** to move axially with respect to the outer shaft, thus causing the yoke of the surgical tip **302** to move axially with respect to the hub, which in turn causes the end effector to actuate (e.g., move between first and second positions). In some cases, manipulation of the grips **352** can be further controlled by a ratcheting mechanism, although that need not always be the case. In some cases, handle **346** can include a rotary knob **350**. Rotary knob **350** can be manipulated to control rotation of the control shaft **304**, thus controlling rotation of the surgical tip **302**.

(48) In some cases, the handle **346** can include a port **348**. Port **348** can act as one or both of an electrical port and a flush port. When used as an electrical port, port **348** can convey electrical current to the inner shaft of the control shaft **304**, which can in turn convey the electrical current through to the end effector **306** of the surgical tip **302** via the yoke. This electrical current can be used for various electrosurgical techniques, including cauterizing. When used as a flush port, port **348** can permit fluid to be flushed through the control shaft **304**, such as to facilitate cleaning and/or sterilization of the control shaft **304**. In some cases, such as when the control shaft **304** is permanently coupled to the handle **346**, the use of a flush port can be especially useful. In some cases, handle **346** can include any number of ports **348**, each of which can act as one or more of an electrical port and a flush port.

(49) FIG. 4 is an exploded schematic side view depicting a surgical instrument tip **400** according to certain aspects of the present disclosure. The surgical tip **400** can include an end effector **406**, a multi-part hub **408**, and a yoke **410**. The multi-part hub **408** can be made of a distal hub **412** and a proximal hub **414** coupled together. The surgical tip **400** can be surgical tip **100** of FIG. 1.

(50) The end effector **406** can include first and second movable portions **416**, **418**. First movable portion **416** can include a pivot hole **454** and a slot **460**. Second movable portion **418** can include a pivot hole **456** and slot **462**. When fully assembled, pivot holes **454**, **456** of the end effector **406** can be aligned coaxially with one another and with pivot **420** of the distal hub **412**. Pivot **420** can be a shaft-like object, such as a rivet, a screw, or a bolt, that fits within pivot holes **454**, **456** to secure the end effector **406** to the distal hub **412** while leaving the first and second movable portions **416**, **418** free to rotate about pivot **420**.

(51) Yoke **410** can be inserted into an inner diameter of hub **408**. Yoke **410** can include a pin **458** that, when the surgical tip **400** is fully assembled, fits within both slots **460**, **462** of the first and second movable portions **416**, **418**. As the pin **458** moves axially within the hub **408**, the pin **458** will move axially with respect to the pivot **420**, which is axially secured within the distal hub **412**. Because of the shape of slots **460**, **462**, axial movement of the pin **458** will thus cause the first and second movable portions **416**, **418** to move between first (e.g., open) and second (e.g., closed) positions.

(52) FIG. 5 is a cross-sectional top view depicting a hub **508** of a surgical instrument tip with the proximal hub **514** and distal hub **512** coupled together according to certain aspects of the present disclosure. The cross-sectional viewpoint of FIG. 5 can be taken along line 5:5 of the hub **408** of FIG. 4. Hub **508** can be hub **408** of FIG. 4 or hub **108** of FIG. 1.

(53) Distal hub **512** can be coupled to proximal hub **514** in any suitable fashion, such as using one or more of a threaded coupling and bonding (e.g., via an epoxy bond). In some cases, proximal hub **514** can overlap a portion **564** of distal hub **512**. Since distal hub **512** can be made of an electrically conductive material, the overlapped portion **564** of the proximal hub **514** over the distal hub **512** can serve to insulate at least that portion **564** of the distal hub **512** from surrounding tissue, thus reducing potential for unwanted electrosurgical actions to non-targeted tissue.

(54) Distal hub **512** can include a pivot **520**, which can be any suitable object, such as a rivet. The pivot **520** can be used to secure an end effector to the distal hub **508**. The proximal hub can couple

to a control shaft, such as through the use of threads **522**.

(55) FIG. **6** is a cross-sectional top view depicting a hub **608** of a surgical instrument tip with the proximal hub **614** and distal hub **612** separate according to certain aspects of the present disclosure. The cross-sectional viewpoint of FIG. **5** can be taken along line 5:5 of the hub **408** of FIG. **4**. Distal hub **612** can be distal hub **512** of FIG. **5**, distal hub **412** of FIG. **4**, or distal hub **112** of FIG. **1**. Proximal hub **614** can be proximal hub **514** of FIG. **5**, proximal hub **414** of FIG. **4**, or proximal hub **114** of FIG. **1**.

(56) In some cases, distal hub **612** can include threads **666** capable of engaging threads **668** of proximal hub **614**. The threads **668** of proximal hub **614** can be separate from or part of the threads used to couple the proximal hub **614** to a control shaft (e.g., threads **522** of FIG. **5**).

(57) In some cases, distal hub **612** and/or proximal hub **614** can include other coupling features used to facilitate coupling the distal hub **612** and the proximal hub **614** together. For example, distal hub **612** and proximal hub **614** can include roughened surfaces (e.g., cross-hatched surfaces) to which epoxy can be applied to provide a more secure bond between distal hub **612** and proximal hub **614** when they are coupled together.

(58) In some cases, distal hub **612** and proximal hub **614** can be coupled together using threads **666**, **668** and epoxy. In an example, epoxy can be applied to one or both of the distal hub **612** and proximal hub **614**, such as at the threads **666**, **668** or elsewhere, after which the distal hub **612** and proximal hub **614** can be screwed together. The threads **666**, **668** can help keep the distal hub **612** and proximal hub **614** in place while the epoxy hardens.

(59) Other techniques for securing the distal hub **612** to the proximal hub **614** can be used. In some cases, distal hub **612** can be removably coupled to proximal hub **614**. In some cases, removable coupling between the distal hub **612** and proximal hub **614** can permit the proximal hub **614** to be replaced with a new proximal hub if it has worn out before the remainder of the surgical tip.

(60) FIG. **7** is a flowchart depicting a process **700** for using a surgical instrument tip according to certain aspects of the present disclosure. At block **702**, a surgical tip can be provided. The surgical tip can be any suitable surgical tip, such as surgical tip **100** of FIG. **1**. In some cases, the surgical tip can be provided from a kit of multiple surgical tips each having a different end effector. In some cases, providing a surgical tip can include providing a sterilized surgical tip in a sterilizing pouch.

(61) At block **704**, the surgical tip can be coupled to a handpiece. The handpiece can include a control shaft coupled (e.g., permanently or removably) to a handle. The surgical tip can be removably coupled to the handpiece. In some cases, coupling the surgical tip to the handpiece can include screwing a yoke of the surgical tip to an inner shaft of the handpiece. In some cases, coupling the surgical tip to the handpiece can include screwing a proximal hub of the surgical tip to an outer shaft of the handpiece. In some cases, coupling the surgical tip to the handpiece can include screwing a yoke of the surgical tip to an inner shaft of the handpiece and screwing a proximal hub of the surgical tip to an outer shaft of the handpiece.

(62) At block **706**, a surgical procedure can be performed using the surgical tip. In some cases, performing a surgical procedure can include passing an electrical current through tissue adjacent the surgical tip using the surgical tip. In some cases, passing electrical current through tissue adjacent the surgical tip can include passing electrical current through targeted tissue adjacent an end effector of the surgical tip and not applying electrical current to non-targeted tissue adjacent the proximal hub of the surgical tip. In some cases, performing a surgical procedure can include actuating an end effector of the surgical tip within a patient. In some cases, performing a surgical procedure can include actuating an end effector of the surgical tip to close one or more movable portions of the end effectors around tissue.

(63) At block **708**, the surgical tip can be removed from the handpiece. Removing the surgical tip from the handpiece can be the opposite of coupling the surgical tip to the handpiece from block **704**, such as unscrewing the surgical tip from the handpiece.

(64) At block **710**, the surgical tip can be sterilized. In some cases, sterilizing the surgical tip can

include cleaning and sterilizing the surgical tip. In some cases, sterilizing the surgical tip can include autoclaving the surgical tip. In some cases, sterilizing the surgical tip can include subjecting the surgical tip to temperatures at or above 121° C. or 135° C. In some cases, sterilizing the surgical tip can include placing the surgical tip in a sterilizing pouch.

(65) At block **712**, the surgical tip, after being sterilized can be coupled to a handpiece, such as the handpiece used at block **704** or another handpiece. Coupling the sterilized surgical tip to a handpiece at block **712** can occur as part of an additional surgical operation distinct from the surgical operation associated with the surgical procedure from block **706**. For example, the surgical procedure from block **706** can be performed on a first patient and coupling the sterilized surgical tip to the handpiece at block **712** can be associated with (e.g., performed in preparation for) a surgical procedure to be performed on a second patient. At block **714**, a surgical procedure can be performed using the sterilized surgical tip from block **712**.

(66) In some cases, instead of proceeding to blocks **712**, **714**, process **700** can loop from block **710** back up to block **702** for subsequent surgical procedures. In some cases, after block **708**, if the surgical tip is deemed to have undergone too many use-and-sterilization cycles (e.g., blocks **702**, **704**, **706**, **708**, **710**) such that the surgical tip is no longer acceptable for use, it can be disposed of or refurbished.

(67) FIG. **8** is a set of schematic side views depicting surgical instrument tips **800**, **802** with flared proximal hubs **814** in closed and open positions according to certain aspects of the present disclosure. Surgical tip **800** can comprise an end effector **806**, a multi-part hub **808**, and a yoke **810**. The yoke **810** is positioned within hub **808** and axially movable with respect to the hub **808** to actuate the end effector **806**. Surgical tip **800** can have an end effector **806** in a first (e.g., closed) position. Surgical tip **802** can be surgical tip **800** with the end effector **806** in a second (e.g., open) position. Surgical tip **800** can be surgical tip **100** of FIG. **1**.

(68) The end effector **806** can include one or more movable portions. As depicted in FIG. **8**, the end effector **806** includes a first movable portion **816** and a second movable portion **818**. The first and second movable portions **816**, **818** are rotatable about a pivot **820** secured to hub **808**. The pivot **820** can be secured to the distal hub **812** of the multi-part hub **808**. Thus, the end effector **806** can be coupled to the distal hub **812** of the multi-part hub **808**. The first and second movable portions **816**, **818** can be movable coupled with respect to the yoke **810** to permit the first and second movable portions **816**, **818** to move between a first position as depicted with respect to surgical tip **800** and a second position as depicted with respect to surgical tip **802** when the yoke **810** is moved axially with respect to the hub **808**. For example, movement of the yoke **810** in direction **840** can cause the first and second movable portions **816**, **818** to move in inward directions **834** to close the first and second movable portions **816**, **818** against one another, while movement of the yoke **810** in direction **842** can cause the first and second movable portions **816**, **818** to move in outward directions **836** to open and separate the first and second movable portions **816**, **818** from one another. Movement of the yoke **810** over distance **844** can represent actuation of the end effector **806** between a first position and a second position. As depicted in FIG. **8**, the first and second movable portions **816**, **818** are jaws of a grasper, although other movable portions can be used.

(69) In some cases, a movable portion of an end effector **806** can have a proximal end that sticks out beyond a circumference of the distal hub **812** when in the first or second position. As depicted in FIG. **8**, the first and second movable portions **816**, **818** of surgical tip **802** have proximal ends **838** that move along a path during actuation of the end effector **806**, which path takes the proximal ends **838** outside of the circumference of the distal hub **812**. The distal hub **812** can include one or more slots through which the proximal ends **838** may pass. The proximal hub **814** can overlap a portion of the distal hub **812** up to or approximately up to such slot(s). In some cases, such as if no slot exists because the proximal ends of the end effector remain within the circumference of the distal hub during actuation, the proximal hub can overlap a portion of the distal hub that extends for up to or approximately up to the pivot **820** or up to or approximately up to the distal end of the

distal hub.

(70) The multi-part hub **808** can comprise a distal hub **812** coupled to a proximal hub **814**. While the distal hub **812** can be made of a metal, such as the same metal used for the end effector **806** and/or the yoke **810**, the proximal hub **814** can be made of an insulator, such as PEEK. The proximal hub **814** can overlap and surround a portion of the distal hub **812**, although that need not always be the case.

(71) The surgical tip **800** can removably couple to a control shaft **804**. The control shaft **804** can include an inner shaft **828** and an outer shaft **826**. For illustrative purposes, the outer shaft **826** is depicted in a partial-cutaway view and the inner shaft **828** is shown in a cutaway side view.

(72) The inner shaft **828** can be axially movable within the outer shaft **826** to facilitate actuation of the end effector **806** via axially moving the yoke **810**. The inner shaft **828**, as well as the yoke **810** and end effector **806**, can be made of electrically conductive materials, such as electrically conductive metal. The outer shaft **826** can be made of an electrically insulating material, such as PEEK. The outer shaft **826** can be made of the same material as the proximal hub **814**.

(73) The proximal hub **814** can removably couple to the outer shaft **826** of the control shaft **804** and the yoke **810** can removably couple to the inner shaft **828** of the control shaft **804**. The proximal hub **814** can include threads **822** (e.g., internal threads) that engage threads **830** (e.g., external threads) of the outer shaft **826**. The yoke **810** can include threads **824** (e.g., external threads) that engage threads **832** (e.g., internal threads) of the inner shaft **828**. For illustrative purposes, threads **822** are depicted in dotted line for surgical tip **800**. In some cases, threads **822**, **824** are coaxial. Likewise, threads **830**, **832** can be coaxial.

(74) As used herein with regards to FIG. **8** and other suitable figures, movement of an inner shaft **828** or movement of a yoke **810** refers to movement of the inner shaft **828** or yoke **810** with respect to the outer shaft **826** or hub **808**, respectively. Because the hub **808** is axially fixed with respect to the outer shaft **826** via a coupling (e.g., threaded coupling) between the outer shaft **826** and the proximal hub **814**, and because the inner shaft **828** is axially fixed to the yoke **810** via a coupling (e.g., threaded coupling), movement of the inner shaft **828** axially with respect to the outer shaft **826** causes the yoke **810** to move axially with respect to hub **808**, thus actuating the end effector to move between first and second positions.

(75) Proximal hub **814** can have a flared profile wherein the outer diameter of the proximal hub **814** changes across its length. In some cases, the proximal end of the proximal hub **814** can have a diameter that matches a diameter of an outer shaft of a control shaft. In some cases, the outer diameter of the proximal hub **814** can increase moving distally from the proximal end of the proximal hub **814**. In some cases, the outer diameter of the proximal hub **814** can increase moving distally from the proximal end of the proximal hub **814** to a maximum diameter, then decrease moving distally towards the distal end of the proximal hub **814**. The profile of the proximal hub **814** can have any suitable contour (e.g., convex or concave) or combination of contours. In some cases, the maximum outer diameter of the proximal hub **814** can have a radial extent from a centerline of the hub **808** that is equal to or approximately equal to a maximum radial extent of the proximal end **838** of the end effector **806**. For example, as depicted with reference to surgical tip **802** of FIG. **8**, the proximal end **838** can extend radially from a centerline of the hub **808** by the same amount or approximately the same amount as the maximum outer diameter of the proximal hub **814**.

(76) The foregoing description of the embodiments, including illustrated embodiments, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or limiting to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art.

(77) As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., “Examples 1-4” is to be understood as “Examples 1, 2, 3, or 4”).

(78) Example 1 is a surgical tip comprising: an end effector movable from a first position to a second position, the end effector having a distal region for contacting tissue and a proximal region; a yoke coupled to the proximal region of the end effector for manipulating the end effector between the first position and the second position, wherein the yoke comprises a threaded proximal end, wherein the yoke and the end effector are made of an electrically conductive material, and wherein the yoke and the end effector are electrically coupled; and a hub positioned around the yoke, wherein the hub comprises a distal hub and a proximal hub, wherein the end effector is coupled to the distal hub at a pivot, wherein the proximal hub is made of a non-conductive material, and wherein the proximal hub comprises a proximal threaded end.

(79) Example 2 is the surgical tip of example(s) 1, wherein the proximal hub is made of Polyetheretherketone.

(80) Example 3 is the surgical tip of example(s) 1 or 2, wherein the proximal hub is coupled to the distal hub by a threaded connection.

(81) Example 4 is the surgical tip of example(s) 1-3, wherein the proximal hub is coupled to the distal hub with an epoxy.

(82) Example 5 is the surgical tip of example(s) 1-4, wherein the threaded proximal end of the yoke and the threaded proximal end of the proximal hub are coaxially threaded.

(83) Example 6 is the surgical tip of example(s) 1-5, wherein the proximal hub surrounds a portion of the distal hub.

(84) Example 7 is the surgical tip of example(s) 1-6, wherein the end effector is selected from the group consisting of scissors, graspers, a punch, and dissectors.

(85) Example 8 is the surgical tip of example(s) 1-7, wherein the distal hub includes a slot for receiving the proximal region of the end effector during movement of the end effector between the first position and the second position, and wherein the proximal hub surrounds a portion of the distal hub located proximally beyond the slot.

(86) Example 9 is the surgical tip of example(s) 1-8, wherein the proximal hub is made of material having a dielectric strength of at or greater than 130 KV/cm.

(87) Example 10 is the surgical tip of example(s) 1-9, wherein the proximal hub has an outer diameter of at or less than 6.0 mm.

(88) Example 11 is the surgical tip of example(s) 1-10, wherein the proximal hub is made of an autoclavable and sterilizable material.

(89) Example 12 is the surgical tip of example(s) 1-11, wherein the proximal hub is made of a material having a glass transition temperature at or above 120° C.

(90) Example 13 is the surgical tip of example(s) 1-12, wherein a length between a distal end of the distal hub and a proximal end of the yoke is at or less than 5 cm.

(91) Example 14 is the surgical tip of example(s) 1-13, wherein a length between a distal end of the end effector and a proximal end of the yoke is at or less than 9 cm.

(92) Example 15 is the surgical tip of example(s) 1-14, wherein the end effector comprises a stationary portion and a movable portion, wherein the stationary portion is fixed with respect to the hub.

(93) Example 16 is the surgical tip of example(s) 1-15, wherein the end effector comprises a first movable portion and a second movable portion, wherein the first movable portion and the second movable portion move with respect to the hub during movement of the end effector between the first position and the second position.

(94) Example 17 is a surgical apparatus comprising: the surgical tip of example(s) 1-16; and a handpiece couplable to the surgical tip, wherein the handpiece comprises an inner shaft movable axially within an outer shaft, wherein the outer shaft is couplable to the proximal hub by the proximal threaded end of the proximal hub, and wherein the inner shaft is couplable to the yoke by the threaded proximal end of the yoke.

(95) Example 18 is the surgical apparatus of example(s) 17, wherein the outer shaft is made of the

same material as the proximal hub.

(96) Example 19 is the surgical apparatus of example(s) 17 or 18, wherein the outer shaft has an outer diameter that is the same size as an outer diameter of the proximal hub.

(97) Example 20 is the surgical apparatus of example(s) 17-19, wherein the handpiece further comprises a cautery pin electrically coupled to the inner shaft to convey electrical signals through the inner shaft and the yoke to the end effector when the surgical tip is coupled to the handpiece.

(98) Example 21 is the surgical apparatus of example(s) 17-20, further comprising an additional surgical tip, wherein the handpiece is couplable to the additional surgical tip after decoupling of the surgical tip from the handpiece, wherein the additional surgical tip comprises an additional end effector that is different from the end effector of the surgical tip.

(99) Example 22 is a method, comprising: providing the surgical tip of example(s) 1-16; coupling the surgical tip to a handpiece, wherein the handpiece comprises an inner shaft movable axially within an outer shaft, and wherein coupling the surgical tip to the handpiece comprises coupling the outer shaft of the handpiece to the proximal hub of the surgical tip by the proximal threaded end of the proximal hub, and coupling the inner shaft of the handpiece to the yoke of the surgical tip by the threaded proximal end of the yoke; decoupling the surgical tip from the handpiece; and autoclaving the surgical tip.

(100) Example 23 is the method of example(s) 22, further comprising: coupling the autoclaved surgical tip to the handpiece or an additional handpiece; and contacting tissue with the end effector of the surgical tip during a surgical procedure.

(101) Example 24 is the method of example(s) 22 or 23, further comprising applying electrical current to tissue through the yoke and the end effector.

(102) Example 25 is the method of example(s) 24, wherein the electrical current conveyed through the yoke is not conveyed through the proximal hub.

Claims

1. A surgical tip comprising: an end effector comprising at least one movable portion movable from a first position to a second position, the end effector having a distal region for contacting tissue and a proximal region; a yoke coupled to the proximal region of the end effector for manipulating the at least one movable portion between the first position and the second position, wherein the yoke comprises a threaded proximal end, wherein the yoke and the end effector are made of an electrically conductive material; a hub positioned around the yoke, wherein the hub comprises: a distal hub made of a conductive material, wherein a proximal portion of the at least one movable portion extends outside of a circumference of the distal hub in the first position or the second position; and a proximal hub externally surrounding a portion of the distal hub, wherein the proximal hub is made of a non-conductive material, and wherein the proximal hub comprises a proximal threaded end and a distal end; and a pivot passing through the distal hub, the pivot coupling the end effector to the distal hub.

2. The surgical tip of claim 1, wherein the at least one movable portion comprises: a pivot hole aligned coaxially with the pivot of the distal hub; and a slot configured to receive a pin of the yoke.

3. The surgical tip of claim 2, wherein the proximal hub surrounds a portion of the distal hub located proximally beyond the slot.

4. The surgical tip of claim 1, wherein at least a portion of the end effector is rotatable around the pivot to move between the first position and the second position.

5. The surgical tip of claim 1, wherein the proximal hub includes a first diameter at the proximal threaded end and a second diameter at the distal end, wherein the first diameter is smaller than the second diameter.

6. The surgical tip of claim 1, wherein the proximal hub is coupled to the distal hub by a threaded connection.

7. The surgical tip of claim 6, wherein the proximal hub is further coupled to the distal hub with an epoxy.
 8. The surgical tip of claim 1, wherein the threaded proximal end of the yoke and the threaded proximal end of the proximal hub are coaxially threaded.
 9. The surgical tip of claim 1, wherein the end effector is selected from the group consisting of scissors, graspers, a punch, and dissectors.
 10. The surgical tip of claim 1, wherein the proximal hub is made of an autoclavable and sterilizable material.
 11. The surgical tip of claim 1, wherein the end effector further comprises a stationary portion fixed with respect to the hub.
 12. The surgical tip of claim 1, wherein the end effector comprises a first movable portion and a second movable portion, wherein the first movable portion and the second movable portion move with respect to the hub during movement of the end effector between the first position and the second position.
 13. A surgical apparatus comprising: the surgical tip of claim 1; and a handpiece couplable to the surgical tip, wherein the handpiece comprises an inner shaft movable axially within an outer shaft, wherein the outer shaft is couplable to the proximal hub by the proximal threaded end of the proximal hub, and wherein the inner shaft is couplable to the yoke by the threaded proximal end of the yoke.
 14. The surgical apparatus of claim 13, wherein the outer shaft has an outer diameter that is the same size as an outer diameter of the proximal hub.
 15. The surgical apparatus of claim 13, wherein the handpiece further comprises a cautery pin electrically coupled to the inner shaft to convey electrical signals through the inner shaft and the yoke to the end effector when the surgical tip is coupled to the handpiece.
 16. The surgical apparatus of claim 13, further comprising an additional surgical tip, wherein the handpiece is couplable to the additional surgical tip after decoupling of the surgical tip from the handpiece, wherein the additional surgical tip comprises an additional end effector that is different from the end effector of the surgical tip.
 17. A method, comprising: providing the surgical tip of claim 1; coupling the surgical tip to a handpiece, wherein the handpiece comprises an inner shaft movable axially within an outer shaft, and wherein coupling the surgical tip to the handpiece comprises coupling the outer shaft of the handpiece to the proximal hub of the surgical tip by the proximal threaded end of the proximal hub, and coupling the inner shaft of the handpiece to the yoke of the surgical tip by the threaded proximal end of the yoke; decoupling the surgical tip from the handpiece; and autoclaving the surgical tip.
 18. The method of claim 17, further comprising: coupling the autoclaved surgical tip to the handpiece or an additional handpiece; and contacting tissue with the end effector of the surgical tip during a surgical procedure.
 19. The method of claim 17, further comprising applying electrical current to tissue through the yoke and the end effector.
 20. The method of claim 19, wherein the electrical current conveyed through the yoke is not conveyed through the proximal hub.
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